Project Proposal

PES University, Department of Computer Science and Engineering
Software Engineering Mini Project, Team 16

1. Project Title & Group

Field	Value
Project Title	Online Internship & Placement Preparation Portal (OIPPP)
Group Number	16
Course	Software Engineering — PES University, Dept. of CSE
Date	26 October 2026

2. Team Profile

#	Name	SRN
1	Deekshith Gowda G	PES1UG24CS142
2	Sharath H G	PES1UG25CS844
3	Guru R Bharadwaj	PES1UG24CS177
4	E S Sukruth	PES1UG25CS160

Every team member shall be involved in all project activities.

3. Project Description

3.1 Overview

OIPPP is a **new, self-contained web application** that brings the two halves of campus placement into one system: **finding internships** and **preparing for them**. Today students track openings across scattered notice boards, spreadsheets and mail threads, while practice for aptitude rounds happens on unrelated third-party sites — so nobody, student or placement officer, can see whether a candidate is actually ready for the roles they are applying to.

OIPPP puts listings, applications, aptitude practice and readiness analytics behind one login, so a student's preparation history and their application history inform each other.

The portal serves three user classes (students, recruiters and placement admins) through a React client and a Node.js + Express API over MongoDB, with JWT sessions and role checks on every route. A student builds a profile, finds and applies to internships, practises timed aptitude tests and sees a readiness view; a recruiter posts listings and moves applicants through a validated status lifecycle; the placement admin approves recruiters, authors tests and reads placement statistics.

3.2 Customer / Stakeholders

We do not yet have a confirmed customer. Prospective customer: the college **Placement Cell**. Interested parties: the placement officer (administrator), recruiting companies, and students.

3.3 Typical Users

User Class	Characteristics	Priority
Student	Pre-final / final year, daily use, low-to-moderate technical expertise. Primary user class.	High
Recruiter	Company representative, occasional use, posts listings and reviews applicants.	High
Placement Admin	College staff, moderates recruiters and listings, consumes placement analytics.	Medium

3.4 Functional Features

The itemized list referenced by Section 4. **What the user will be able to do:**

ID	Functional Feature	Primary User
F1	Register and log in as student, recruiter or admin	All
F2	Role-based access to permitted screens and data only	All
F3	Create and maintain a student profile — education, skills, resume	Student
F4	Administer users — approve recruiters, deactivate accounts	Admin
F5	Post, edit and close internship listings	Recruiter
F6	Search, filter and sort internship listings	Student
F7	Bookmark listings; listings expire automatically after the deadline	Student
F8	Receive skill-matched internship recommendations	Student
F9	Apply to an internship with resume and cover note	Student
F10	Track application status through its lifecycle	Student
F11	Review applicants — shortlist, reject with reason	Recruiter
F12	Receive notifications on every application status change	Student
F13	Author aptitude / technical tests from a question bank	Admin
F14	Attempt a timed test with answers auto-saved	Student
F15	Receive an auto-evaluated score with a per-topic breakdown	Student
F16	View progress and analytics dashboards	Student, Admin

3.5 Out of Scope (v1)

Mobile applications, payments, video interviews, résumé parsing by ML, and integration with external job boards.

4. Plan of Work and Product Ownership

4.1 Short-Term Plan (next few weeks)

Week	Planned Work
1	Proposal finalised; repository, branching and toolchain set up
2–3	SRS written; use-case and ER diagrams; API contract agreed
4	Schemas implemented; authentication working end-to-end; UI shell
5–6	Listings and applications features built against the agreed contract

4.2 Who Should Do What

Ownership is by **functional feature**, per the guidelines. Full task-level breakdown and live status live in the README and the GitHub project board.

Member	Functional Features Owned	Qualitative Property Contributed
Member 1 — Deekshith Gowda G	F1, F2, F3, F4	**Confidentiality** — passwords stored hashed, role boundaries enforced on every route, a resume readable only by its owner and recruiters they applied to
Member 2 — Sharath H G	F5, F6, F7, F8	**Performance** — a filtered listing search returns in under 2 seconds against 5,000+ seeded listings
Member 3 — Guru R Bharadwaj	F9, F10, F11, F12	**Reliability** — no application is lost or duplicated; every status transition is validated and audit-logged
Member 4 — E S Sukruth	F13, F14, F15, F16	**Usability & fairness** — an in-progress timed test survives a page refresh or dropped connection with answers and clock intact

Load is balanced at four functional features and one qualitative property per member. User interface, graphic design, database interaction and unit testing are **not** counted as functional features — each member performs those for the features they own.

5. Technology Stack

Layer	Choice
Frontend	React
Backend	Node.js + Express
Database	MongoDB
Auth	JWT + bcrypt
Analytics / scoring service	Python (optional service, per the problem statement's *Python / JavaScript* stack)
Testing	Jest + Supertest